

ATTACHMENT 5

REQUIREMENT DOCUMENT

NEXT GENERATION SURFACE SEARCH RADAR (NXSSR) SOFTWARE DEVELOPMENT KIT (SDK) AND AUTOMATIC RADAR PLOTTING AID (ARPA) APPLICATION FOR THE UNITED STATES COAST GUARD (USCG) COMMAND, CONTROL, COMMUNICATIONS, COMPUTERS, CYBER, AND INTELLIGENCE SERVICE CENTER (C5ISC)

1 GENERAL

1.4 BACKGROUND

The Coast Guard (CG) uses Surface Search Radar (SSR) to provide operators with an advanced system for collision avoidance, safe navigation, and enhanced situational awareness of the maritime environment. There are currently five (5) separate radars (AN/SPS-50, AN/SPS-78, AN/SPS-79, AN/SPS-73, and Terma Scanter 2001) in use across 14 cutter platforms and 113 Vessel Traffic Service (VTS) remote sites. In May 2019, the Coast Guard was notified that the FAR 2XXX series (which includes the AN/SPS-50 radar) was in final production and end-of-support. The AN/SPS-78 and AN/SPS-79 radars precede the AN/SPS-50, are significantly degraded, and overdue for recapitalization. Ashore, the VTS Terma Scanter 2001 Radars are end-of-support and require replacement. The AN/SPS-73 radar is beyond its service life and needs replacement.

The current state SSRs utilize several different interfaces for afloat Command and Control (C2) and ashore VTS systems. For afloat applications, the AN/SPS-50 radar is a proprietary Furuno system that interfaces with CG Electronic Chart Display and Information System (ECDIS) and CG Electronic Charting System (ECS) by use of a Buffalo Computer Graphics integration solution, NAVNET IMO, for radar overlay. The AN/SPS-78 and AN/SPS-79 radars utilize Furuno turning units and arrays with SSR Engineering Radar Processors (PCRP) which interface to the C2 Systems, SeaWatch and Sea Commander. The PCRP interface via COMRIC which feeds the C2 system clients running the current state Command, Display and Control (C) – Automatic Radar Plotting Aid (C-ARPA) Application.

For ashore, the Terma Scanter 2001 and AN/SPS-73 radars interface with the VTS Ports and Waterways Safety System (PAWSS), a Lockheed Martin Corporation core software product (MTM-300). PAWSS interfaces with TERMA Scanter 2001 radars via a Curtis Wright Dual channel radar interface card that processes video, trigger, and azimuth data that translates into centroids and track data. The AN/SPS-73 radars interface with PAWSS by use of SSR Engineering PCRP 201A radar processors. Furuno radar data is processed into track data and transmitted to the PAWSS core software.

1.4 SCOPE

The Coast Guard (CG) seeks to procure a Software Development Kit (SDK), a correlating Automatic Radar Plotting Aid (ARPA) Application, and necessary enterprise licensing. The SDK must be a customizable radar integration kit to leverage across the scalable, IMO-certified Surface

Search Radar (SSR) system. The ARPA application must interface with C2 systems by use of a Local Area Network (LAN) and enable video data, target data, and control functions providing Plan Position Indicator (PPI) duplication and functionality to include AIS track correlation for all CG Enterprise Electronic Navigation System solutions to include but not limited to, CG-ECS, CG-ECDIS, SeaWatch, and Sea Commander.

There are five (5) distinct tasks associated with this effort:

- Task A - Software Development Kit (SDK)
- Task B - SDK Documentation
- Task C - SDK Enterprise Licensing
- Task D - Automatic Radar Plotting Aid (ARPA) Application
- Task E - ARPA Application Enterprise Licensing

1.4 APPLICABLE DOCUMENTS

1.4.1 Compliance Documents

The Contractor must adhere to all policies and procedures established by the Government, Safety Code, or Industry Standard. This is not a complete list of all references; therefore, the Contractor is required to meet all standard requirements /specifications, whether the reference document is listed or not. The documents listed below provide specifications, standards, or guidelines that must be complied with to meet established requirements:

- a. Coast Guard Next Generation Surface Search Radar (NXSSR) System Requirement Document (SRD)

1.4 GOVERNMENT TERMS AND DEFINITIONS

Acronym	Definition
API	Application Programming Interface
ARPA	Automatic Radar Plotting Aid
C2	Command & Control
C5ISC	Command, Control, Communications, Computers, Cyber and Intelligence Service Center
C-ARPA	Command, Display & Control System / Automatic Radar Plotting Aid
CG	Coast Guard
COR	Contacting Officer Representative
DISA	Defense Information System Agency
ECDIS	Electronic Chart Display and Information System
ECS	Electronic Charting System
IDIQ	Indefinite-Delivery Indefinite-Quantity
KO	Contracting Officer
KS	Contract Specialist
LAN	Local Area Network
MVP	Minimum Viable Product
NXSSR	Next Generation Surface Search Radar

PAWSS	Ports and Waterways Safety System
PCRP	SSR Engineering Radar Processors
PPI	Plan Position Indicator
SBOM	Software Bill of Materials
SDK	Software Development Kit
SSR	Surface Search Radar
SVD	Software Version Document
USCG	United States Coast Guard
VTS	Vessel Traffic Service

2 REQUIREMENTS

2.1 Task A - Software Development Kit (SDK)

2.1.1 The contractor must provide an SDK which exposes video data, target data, and control functions for inclusion in other products developed by the Government or other contractors, including exposing the functions via Application Programming Interface (API).

2.1.2 The functions provided by the SDK must deliver control of the system based on prioritization by the Government and identified through Attachment 2 - NXSSR Technical Specification.

2.1.3 Software installation procedures for Windows and Linux must be contractor-provided, to include requirements of the target host/platform which must be met to allow for installation.

2.1.4 The Contractor must provide the SDK to the Government in Contractor's preferred format no later than thirty (30) calendar days after task order award.

2.2 Task B - SDK Documentation

2.2.1 The contractor must provide manual(s) explaining each method of the SDK, with examples of implementation for the methods, written to the knowledge level of a software developer.

2.2.1.1 API calls must be described in adequate detail to include all available methods and possible responses, including any returned error conditions and success codes/messages. API examples for explanation should be documented at the level of a software developer.

2.2.3 The SDK documentation must be provided in a digital markdown format for use in a Git repository.

2.2.4 The Contractor must provide the SDK documentation to the Government in Contractor's preferred format no later than thirty (30) calendar days after task order award.

2.3 Task C – SDK Enterprise License

2.3.1 The contractor must provide an enterprise license for the SDK to include unlimited usage for CG Enterprise to include Government Contractors supporting the CG.

2.4 Task D – Automatic Radar Plotting Aid (ARPA) Application

2.4.1 The contractor must develop an ARPA Application based on Software Development Kit (SDK) that uses exposed functions via Application Programming Interface (API) with Windows 11 and Linux variants.

2.4.2 The ARPA Application must allow direct radar interface via Local Area Network (LAN) with existing C2 Systems providing radar video (PPI), target data, and control functions.

2.4.3 ARPA Application functionality must include AIS track correlation with radar targets.

2.4.4 Each Minimum Viable Product (MVP) must include a Software Bill of Materials (SBOM) for each component.

2.4.5 Initial ARPA Application delivery and any future deliveries throughout the life of the order must include Git repository updates to include source code with the associated change history including a Software Version Document (SVD) noting all changes from the previous software version.

2.4.6 The Contractor must provide the ARPA Application to the Government no later than one hundred and eighty (180) calendar days after task order award.

2.5 Task E – ARPA Application Enterprise License

2.5.1 The contractor developed ARPA Application must be provided with an unlimited license for use to include rights to use, modify, reproduce, perform, display, release, or disclose technical data in whole or in part, in any manner, and for any purpose whatsoever, and to have or authorize others to do so. This includes modifications to the ARPA Application for further development, as required.

3.0 OTHER APPLICATION CONDITIONS

3.1 SECURITY

3.1.1 Contractor personnel working on-site at Government facilities must comply with all installation security requirements and all security regulations and directives for this IDIQ Contract (i.e., security and safety, visit request, badging, escorted/unescorted, training, day-to-day requirements, etc.).

3.1.2 The contractor will ensure personnel hold U.S. citizenship and must provide a list of employees assigned functions for this IDIQ contract to the COR. The contractor will advise the COR immediately upon change of personnel associated with this IDIQ contract.

3.1.3 Contractor must submit a Visitor Access Requests (VAR) / VAL reflecting adjudicated investigation data populated from the Defense Counterintelligence & Security Agency (DCSA) enterprise-wide solution for personnel security, suitability, and credentialing management for DoD military, civilian, and contractors.

3.1.4 Contractor personnel working under this IDIQ Contract, at a minimum, must have a favorable Tier 1 investigation for physical and logical access to obtain a DoD Common Access Card (CAC) in accordance with the COMDTINST 5500.18, Mission Partner, Identity Credential and Access Management (MP ICAM). Public Trust (T2 or T4) investigation is based on assigned duties. Any access to classified National Security Information, and / or Classified Workspaces (CWS) contractor personnel must meet the appropriate level of clearance for Tier 3 investigation standards.

3.1.5 All hardware, software, and services provided must be compliant in accordance with 140-01 Information Technology Systems Security and the DHS Sensitive Systems Handbooks 4300A for Sensitive but Unclassified (SBU).

3.1.6 Physical Access

3.1.6.1 If a CAC is not required per COMDTINST 5500.18, then the provisions of COMDTINST 5532.1, Physical Access Control will be adhered to including a NCIC-III criminal history inquiry, favorable unescorted access determination, and locally produced access credential issuance (i.e. visitor's pass, temporary badge, etc.) must be followed.

3.1.6.2 Personnel who are granted physical access, without determination of a favorable fitness, must be accompanied by an Authorized Escort Authority as outlined per COMDTINST 5532.1, Chapter 9.D. The escort requirement is mandated for the duration of the individual's visitation. In the event the classified work needs to be conducted in the space identified in this IDIQ Contract, the Contractor understands all work will be placed on hold and will continue once the CWS has been cleared by security personnel.

3.1.6.3 Contractor must adhere to the access control requirements of the installation for site visits. Contractor's employees may be exposed to For Official Use Only (FOUO) while visiting the Government sites, which require the contractor's employees to sign the DHS Non-Disclosure Agreement (NDA), Form 11000-6.

3.2 PERIOD OF PERFORMANCE

The period of performance for this task order one year from date of award.

3.3 SECTION 508 COMPLIANCE

3.3.1 Section 508 Requirements

Section 508 of the Rehabilitation Act (classified to [29 U.S.C. § 794d](#)) requires that when Federal agencies develop, procure, maintain, or use information and communications technology (ICT), it shall be accessible to people with disabilities. Federal employees and members of the public with disabilities must be afforded access to and use of information and data comparable to that of Federal employees and members of the public without disabilities.

All products, platforms and services delivered as part of this work statement that, by definition, are deemed ICT shall conform to the revised regulatory implementation of Section 508 Standards, which are located at 36 C.F.R. § 1194.1 & Appendixes A, C & D, and available at <https://www.ecfr.gov/cgi-bin/text-idx?SID=e1c6735e25593339a9db63534259d8ec&mc=true&node=pt36.3.1194&rgn=div5>. In the revised regulation, ICT replaced the term electronic and information technology (EIT) used in the original 508 standards. ICT includes IT and other equipment.

Exceptions for this work statement have been determined by DHS and only the exceptions described herein may be applied. Any request for additional exceptions shall be sent to the Contracting Officer and a determination will be made according to DHS Directive 139-05, Office of Accessible Systems and Technology, dated November 12, 2018 and DHS Instruction 139-05-001, Managing the Accessible Systems and Technology Program, dated November 20, 2018, or any successor publication.

3.3.1.2 Section 508 Requirements for Technology Products

Section 508 applicability to Information and Communications Technology (ICT): Software Development

Applicable Exception: National Security Authorization #: EXC0001554

Applicable Functional Performance Criteria: Does not apply

Applicable 508 requirements for electronic content features and components: Does not apply

Applicable 508 requirements for software features and components: Does not apply

Applicable 508 requirements for hardware features and components: Does not apply

Applicable 508 requirements for support services and documentation requirements: Does not apply

3.3.1.3 Section 508 Requirements for Technology Products

Section 508 applicability to Information and Communications Technology (ICT): System Integration Installation

Applicable Exception: National Security Authorization #: EXC0001554

Applicable Functional Performance Criteria: Does not apply

Applicable 508 requirements for electronic content features and components: Does not apply

Applicable 508 requirements for software features and components: Does not apply

Applicable 508 requirements for hardware features and components: Does not apply

Applicable 508 requirements for support services and documentation requirements: Does not apply

3.3.1.4 Section 508 Requirements for Technology Services

3.3.1.4.1 When developing or modifying ICT, the Contractor is required to validate ICT deliverables for conformance to the applicable Section 508 requirements. Validation shall occur on a frequency that ensures Section 508 requirements is evaluated within each iteration and release that contains user interface functionality.

3.3.1.4.2 When modifying, installing, configuring or integrating commercially available or government-owned ICT, the Contractor shall not reduce the original ICT Item's level of Section 508 conformance.

3.3.1.4.3 When developing or modifying electronic documents and forms provided in a Microsoft Office or Adobe PDF format, the Contractor shall demonstrate conformance to the applicable to the applicable Section 508 standards (including WCAG Level A and AA Level 2.0 Success Criteria) by conducting testing using the test methods published under "Accessibility Tests for Documents" at <https://www.dhs.gov/compliance-test-processes>.

3.3.1.4.4 When developing or modifying ICT deliverables that contain the ability to automatically generate electronic documents and forms in Microsoft Office and Adobe formats, or when the capability is provided to enable end users to design and author web based electronic content (i.e. surveys, dashboards, charts, data visualizations, etc.), the Contractor shall demonstrate the ability to ensure these outputs conform to the applicable Section 508 standards (including WCAG 2.0 Level A and AA Success Criteria). The Contractor shall demonstrate conformance by conducting testing and reporting test results based on representative sample outputs. For outputs produced as Microsoft Office and Adobe PDF file formats, the Contractor shall use the test methods published under "Accessibility Tests for Documents", which are published at <https://www.dhs.gov/compliance-test-processes>. For outputs produced as web based electronic content, the Contractor shall use the DHS Trusted Tester for Web Methodology Version 5.0, or successor versions. This methodology is published at <https://www.dhs.gov/trusted-tester>

3.3.1.4.5 Contractor personnel shall possess the knowledge, skills and abilities necessary to address the accessibility requirements in this work statement.

3.3.1.5 Section 508 Deliverables

3.3.1.5.1 Section 508 Test Plans: When developing or modifying ICT pursuant to this contract, the Contractor shall provide a detailed Section 508 Conformance Test Plan. The Test Plan shall describe the scope of components that will be tested, an explanation of the test process that will be used, when testing will be conducted during the project development life cycle, who will conduct the testing, how test results will be reported, and any key assumptions.

3.3.1.5.2 Section 508 Test Results: When developing or modifying ICT pursuant to this contract, the Contractor shall provide test results in accordance with the Section 508 Requirements for Technology Services provided in this solicitation.

3.3.1.5.3 Section 508 Accessibility Conformance Reports: For each ICT item offered through this contract (including commercially available products, and solutions consisting of ICT that are developed or modified pursuant to this contract), the Offeror shall provide an Accessibility Conformance Report (ACR) to document conformance claims against the applicable Section 508 standards. The ACR shall be based on the Voluntary Product Accessibility Template Version 2.0 508 (or successor versions). The template can be found at <https://www.itic.org/policy/accessibility/vpat>. Each ACR shall be completed by following all of the instructions provided in the template, including an explanation of the validation method used as a basis for the conformance claims in the report.

3.3.1.6 Other Section 508 Documentation: The following documentation shall be provided upon request for ICT items offered through this contract:

3.3.1.6.1 Documentation of features provided to help achieve accessibility and usability for people with disabilities.

3.3.1.6.2 Documentation on how to configure and install the ICT Item to support accessibility.

3.3.1.6.3 Documentation of core functions that cannot be accessed by persons with disabilities.

3.3.1.6.4 Documentation of remediation plans to address non-conformance to the Section 508 standards

3.4 TASK ORDER ADMINISTRATION

3.4.1 Task Order Administration/Management

Notwithstanding the Contractor's responsibility for total management during the performance of the Task Order, the administration of the Task Order will require close coordination between the Government and the Contractor. The U.S. Coast Guard, CG-C5I-C32 will perform all Task Order administration. The individuals listed below will be the Government points of contact during the performance period.

Contracting Office: U.S. Coast Guard
CG-C5I-C32
4000 Coast Guard Blvd.
Portsmouth, VA 23703-2199

3.4.2 Task Order Contracting Officer (KO)

Christina M. Grimstead
Telephone #: (703) 417-5611
E-Mail: Christina.M.Grimstead@uscg.mil

3.4.3 Task Order Administrator (Contract Specialist/KS)

Amanda Barraclough
Telephone #: (703) 417-5612
E-Mail: Amanda.R.Barraclough2@uscg.mil

3.4.4 Task Order Contracting Officer's Representative (COR)

A COR Appointment Memorandum will be provided to the Contractor and will provide the name, telephone number and e-mail address of the COR and set forth the authorities and limitations of the COR under the Task Order.

3.5 INVOICING

3.5.1 Invoicing Schedule

The Contractor may submit invoices for tasks/deliverables that are completed, received, inspected, and accepted by the Government.

3.5.2 Invoicing and Payment Information

The Contractor must comply with the Invoicing and Payment Information in the IDIQ Contract.

3.6 TASK ORDER CLAUSES

3.6.1 Indefinite-Delivery Indefinite-Quantity (IDIQ) Contract Clauses

This Task Order is subject to the clauses in the IDIQ Contract; the clauses are incorporated by reference.

3.6.2 Additional Clauses

The following clauses are applicable to this Task Order.

HSAR Clause 3052.242-72 Contracting Officer's Representative. (DEC 2003)

(a) The Contracting Officer may designate Government personnel to act as the Contracting Officer's Representative (COR) to perform functions under the contract such as review or inspection and acceptance of supplies, services, including construction, and other functions of a technical nature. The Contracting Officer will provide a written notice of such designation to the Contractor within five working days after contract award or for construction, not less than five working days prior to giving the contractor the notice to proceed. The designation letter will set forth the authorities and limitations of the COR under the contract.

(b) The Contracting Officer cannot authorize the COR or any other representative to sign documents, such as contracts, contract modifications, etc., that require the signature of the Contracting Officer.

(End of Clause)